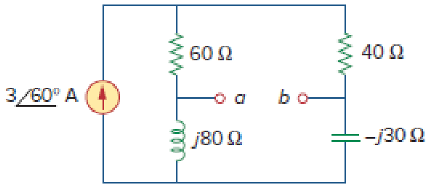
Chapter 10, Problem 64P

Problem

For the circuit shown in Fig. 10.107, find the Norton equivalent circuit at terminals *a*-*b*.

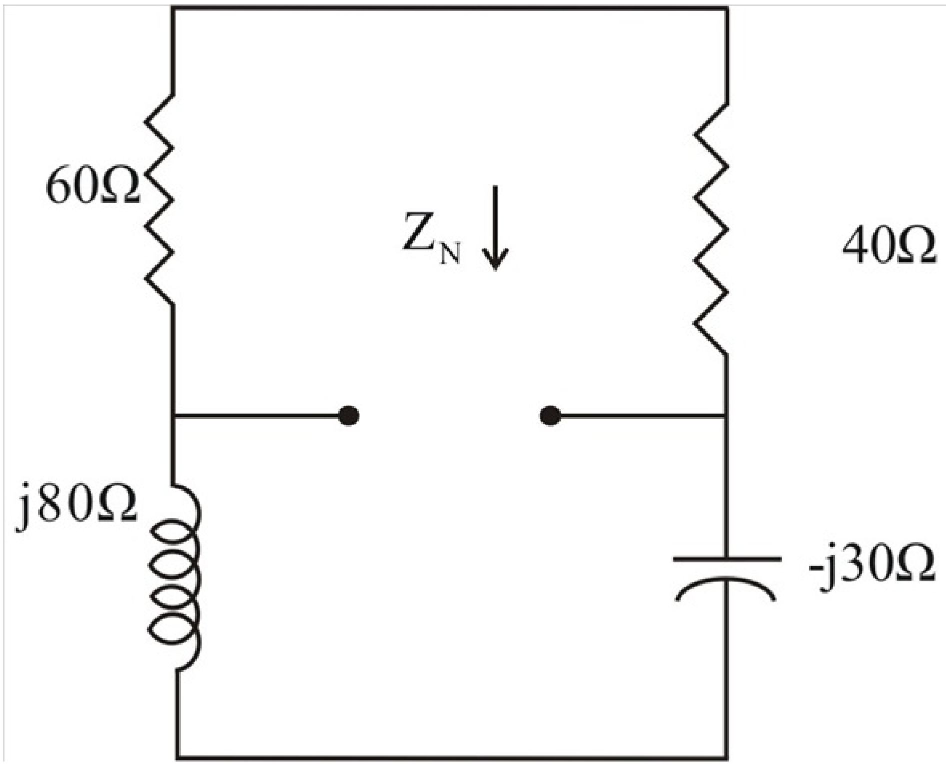
Figure 10.107:



Step-by-step solution

Step 1 of 3

ZN is obtained from the circuit in fig(a),

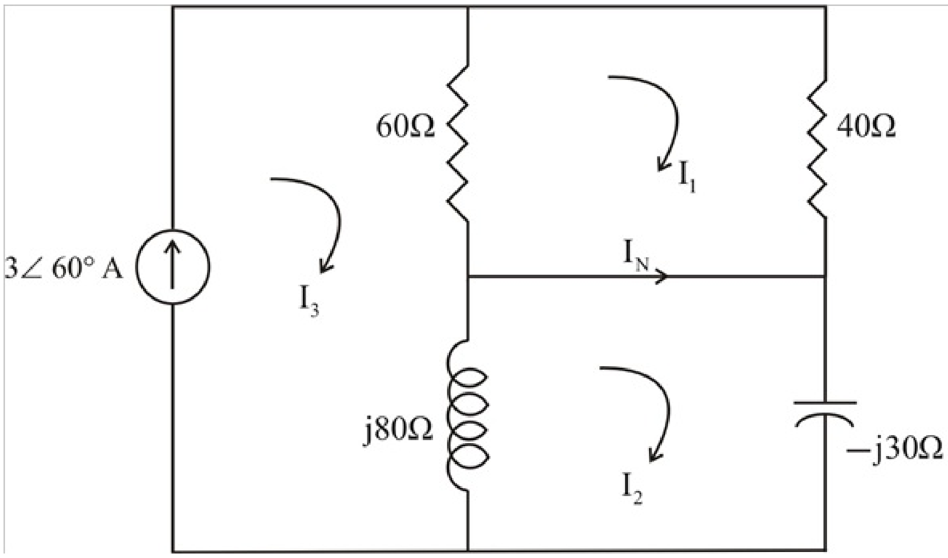


$$Z_N = (60+40) \parallel (j80 - j30) = 100 \parallel 50j = \frac{(100)(j50)}{100 + j50}$$

$$Z_N = 20 + j40 = 44.72\angle 63.43^\circ \ \Omega$$

Step 2 of 3

To find IN, Consider the circuit in fig(b),



$$I_S = 3\angle 60°$$

Step 3 of 3

From Mesh 1,

$$100 I_1 - 60 I_S = 0$$

$$I_1 = 1.8\angle 60°$$

From Mesh 2,

$$(j80 - j30) I_2 - j80 I_3 = 0$$

$$I_2 = 4.8\angle 60°$$

$$I_N = I_2 - I_1 = 3\angle 60° \text{ A} \quad .$$